

DARIUS BIGBEE

GRAPHICS PROGRAMMER | GAME ENGINE DEVELOPER

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[Engine Repository](#) | [GitHub](#) | [Portfolio](#) | [LinkedIn](#)

SUMMARY

Computer Graphics Engineer with an M.S. in Computer Graphics Technology from Purdue University and experience in graphics programming, real-time rendering, virtual reality, game development, and technical instruction. Currently building a custom game engine with Vulkan rendering, ECS architecture, PhysX integration, asset pipelines, and advanced rendering systems. Focused on engine architecture, rendering optimization, and scalable graphics technologies.

PROJECTS

Custom Game Engine | Graphics Programmer & Engine Developer

- Built a Vulkan-based renderer with robust GPU resource management and modern rendering workflows.
- Implemented physically based rendering, shadow mapping, and forward and deferred rendering pipelines.
- Designed a render graph architecture to schedule render passes and manage dependencies efficiently.
- Implemented Multi-Draw Indirect rendering to reduce CPU overhead and increase rendering throughput.
- Developed batching systems for lines, quads, and meshes to minimize draw calls and improve performance.
- Created streamlined rendering paths for debug visualization and 2D rendering.
- Designed and implemented an Entity Component System and hierarchical scene graph for modular scene management.
- Integrated NVIDIA PhysX to support collision detection and real-time physics simulation.
- Developed a skeletal animation system for animated characters and scene objects.
- Created an Inspector system to support runtime editing, debugging, and engine tooling.
- Implemented shader, asset, and resource management systems to streamline engine workflows.
- Used CMake for build automation and Git/GitHub for version control and collaborative development.

Interactive 3D Modeling in Virtual Reality | Master's Thesis

- Developed immersive virtual reality workflows for interactive 3D modeling and content creation.

Crisis Response Leadership Training (CRLT) | Lead Programmer

- Contributed to an interactive simulation platform designed to support emergency response leadership training.

TECHNICAL SKILLS

Languages: C++, C#, GLSL

Graphics: Vulkan, OpenGL, PBR, Forward Rendering, Deferred Rendering, Shadow Mapping, Multi-Draw Indirect

Engine Development: ECS, Scene Graphs, Render Graphs, PhysX, Animation Systems, Asset Management, Shader Management

Tools: Git, GitHub, CMake, Unreal Engine, Maya, ZBrush

PROFESSIONAL EXPERIENCE

Teacher | Right Steps Child Development | 2021–2025

- Managed classroom operations, instructional delivery, and daily planning.
- Strengthened leadership, communication, and project coordination skills in a fast-paced educational environment.

Teacher Assistant / Co-Teacher | Brookshire Learning Center | 2026

- Supervised and instructed children ages 2–5 in a structured classroom setting.
- Served as acting lead teacher, overseeing classroom instruction and daily operations.
- Supported lesson planning, classroom management, and student engagement.

3D Artist | Purdue Polytechnic Institute | 2019

- Created 3D assets for virtual reality applications in an academic research environment.

Instructor & Teaching Assistant | Purdue Polytechnic Institute | 2016–2018

- Taught computer graphics concepts and Unreal Engine fundamentals to students.
- Led workshops and delivered hands-on technical instruction in graphics-related coursework.

EDUCATION

Master of Science, Computer Graphics Technology

Purdue University

Bachelor of Science, Computer Graphics Technology

Purdue University

ADDITIONAL ACHIEVEMENTS

- 2nd Place, Altice & Infosys Innovation Hackathon